

Lab Report

Risk-Aware Payment Gateway: Five-Week Implementation Plan

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Five-Week Implementation Routine

Risk-Aware Payment Gateway

The project is a payment gateway that sits between a merchant checkout and a payment processor. Before a payment is authorised, a risk engine scores it, and the gateway allows it, asks for step-up verification (OTP), or holds it for manual review. Money is never blocked automatically for high-risk cases; a human reviewer decides. The work below will be done from scratch, one week at a time.

Week	Focus	Key tasks	Deliverable and exit gate
1	**Planning and design**	• Requirements, actors (customer, merchant, reviewer) and payment flow • Architecture, REST API contract, database schema (merchants, payments, ledger) • Repo, environment, project skeleton, merchant API-key authentication • Payment state machine: created, authorised, captured, failed, refunded	Approved design document and working skeleton; schema migrates cleanly.
2	**Core payment gateway**	• Create-payment, capture, refund and status endpoints • Idempotency keys so retries never double-charge • Sandbox processor integration and signed webhooks • Double-entry ledger and transaction history	A test payment completes end to end in sandbox; repeated requests create one charge.
3	**Risk engine**	• Synthetic labelled transactions (scam, account takeover, mule, cash-out patterns) • Past-only features: velocity, amount deviation, new device/receiver/location • Classifier plus anomaly score; graph signal for linked accounts • Score inside the payment flow; threshold bands in a config file; reasons shown	Held-out AUC ≥ 0.95; scoring p95 latency under 200 ms; thresholds change without retraining.
4	**Dashboard and security**	• Merchant dashboard: payments, refunds, disputes • Reviewer queue with approve/decline and audit log • OTP step-up, HTTPS, rate limiting, input validation, secret handling • Fairness check across customer segments	Reviewer can act on a held payment in a few clicks; invalid input returns 4xx, never 500.
5	**Testing, deployment, documentation**	• Unit, integration, security and load tests • CI pipeline and cloud deployment • Evaluation against a simple rule baseline • README, API docs, final report and demo rehearsal	CI green; live URL processes a sandbox payment with a risk decision; demo runs end to end.

Note. Targets above are acceptance thresholds for the plan, not results. Real card data is out of scope; only sandbox payments and synthetic risk data are used.